House Price Prediction Using Machine Learning

Project Overview

Objective

- Predict house prices using property features.
- Analyze factors influencing property value.
- Compare machine learning models for prediction accuracy.
- Support data-driven decision-making in real estate.

Dataset

- Housing Prices Dataset (Kaggle)
- Target Variable: Price
- Features: Area, Bedrooms, Bathrooms, Stories, Parking, Furnishing Status, etc.

Data Preprocessing

Data Preparation Steps

- Loaded dataset using Pandas.
- Checked dataset shape and structure.
- Identified target and feature variables.
- Checked missing values and duplicates.
- Converted categorical variables into numerical format using encoding.
- Prepared cleaned dataset for machine learning.

Libraries Used

- Pandas
- NumPy
- Scikit-Learn

Model Development

Machine Learning Models

Linear Regression

- Simple regression algorithm.
- Establishes linear relationship between features and house price.

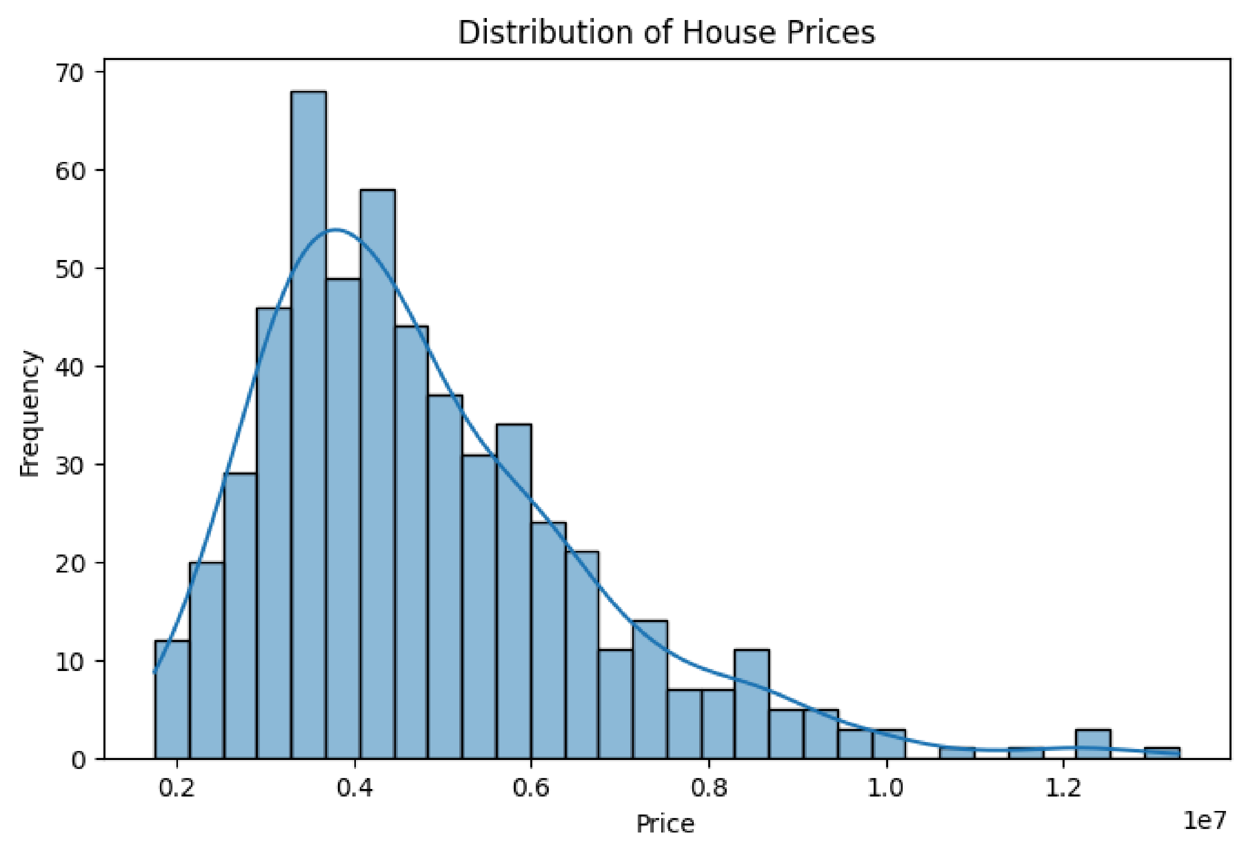
Random Forest Regressor

- Ensemble learning method.
- Captures complex and non-linear relationships.
- Generally provides higher prediction accuracy.

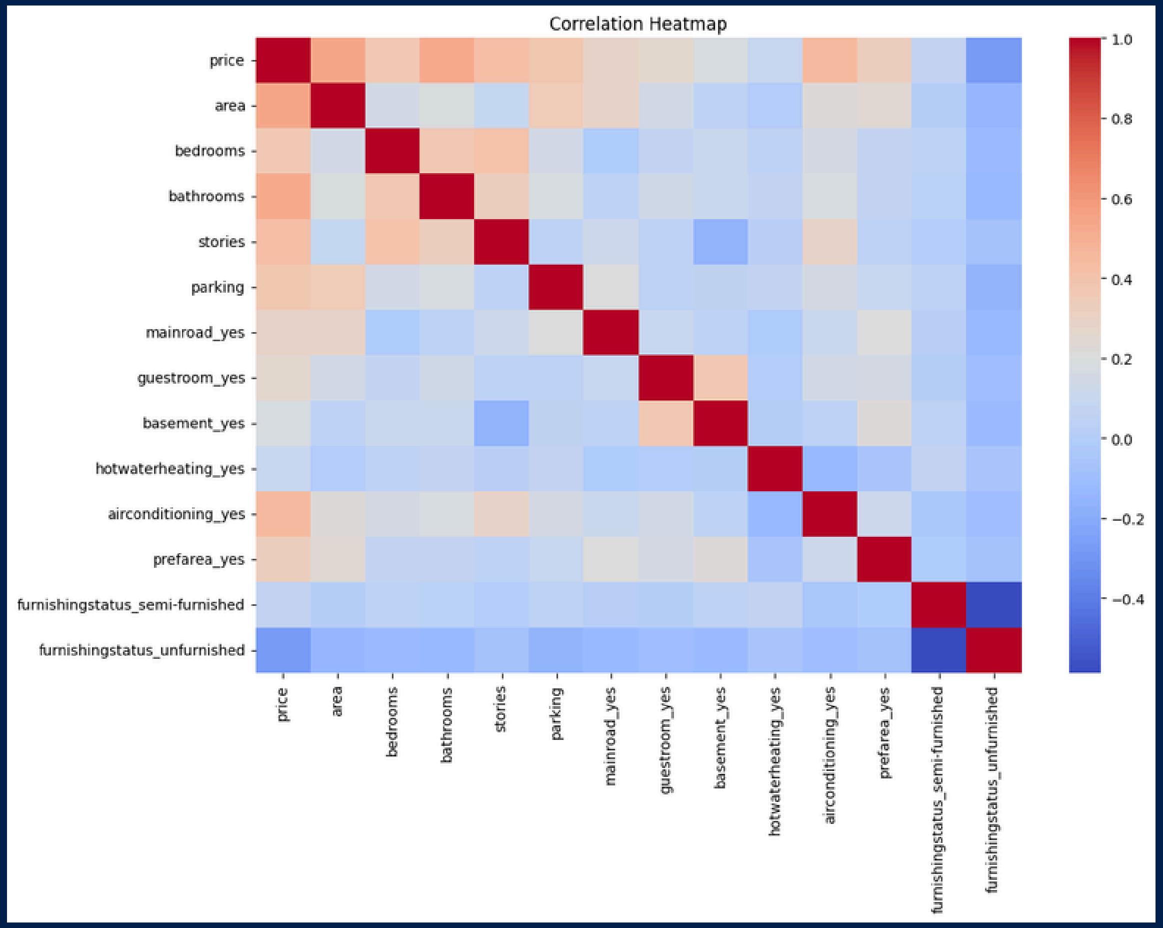
Data Split

- Training Data: 80%
- Testing Data: 20%

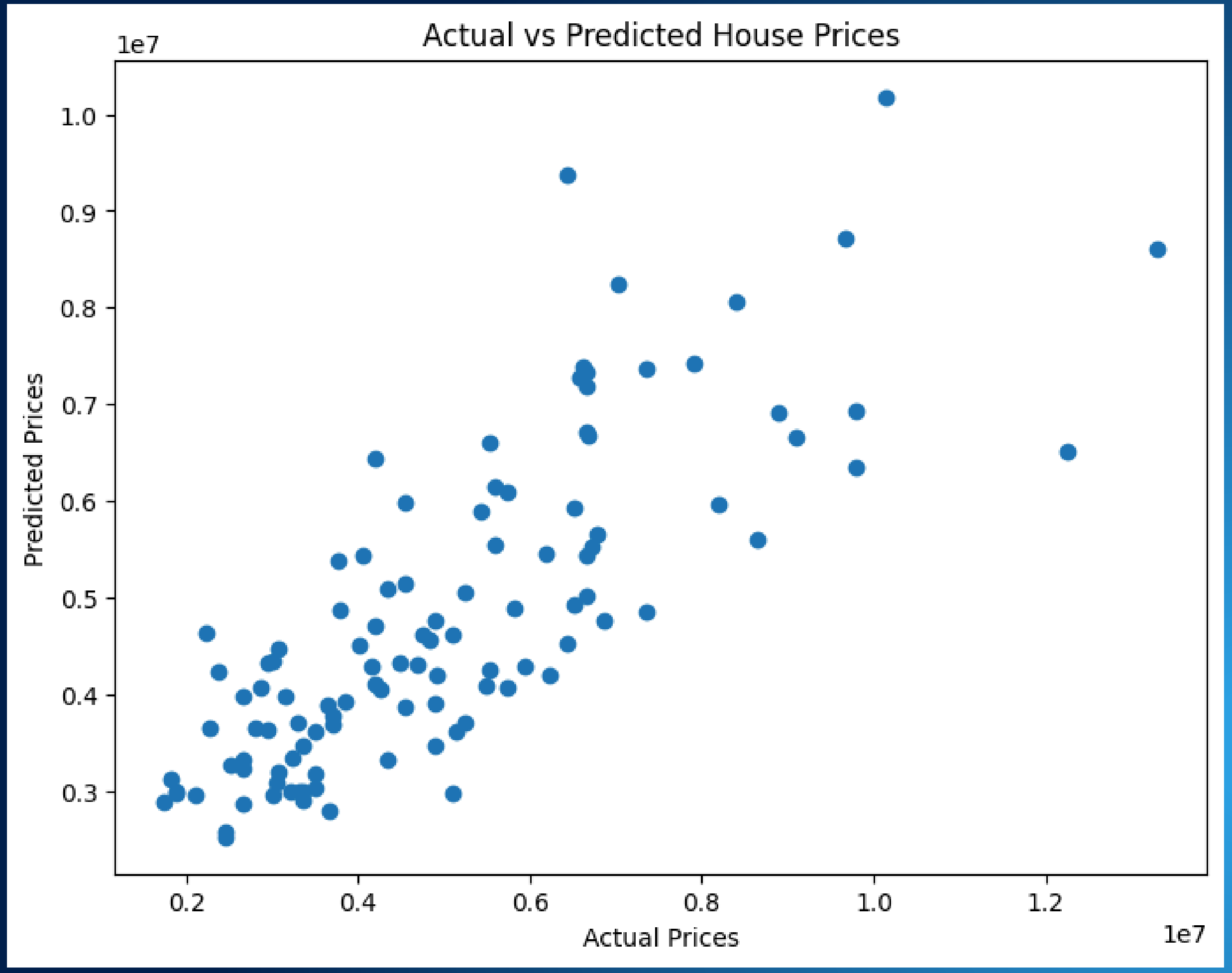
Data Visualization



House Price Distribution Histogram



Correlation Heatmap



Actual vs Predicted Price Scatter Plot

Purpose

- Understand price distribution.
- Identify important features.
- Evaluate prediction performance.

Results & Findings

Key Findings

- Area strongly influences house prices.
- Number of bathrooms and stories significantly affect value.
- Parking availability contributes positively to price.
- Location-related attributes impact market valuation.
- Random Forest performed better than Linear Regression.

Performance Metrics

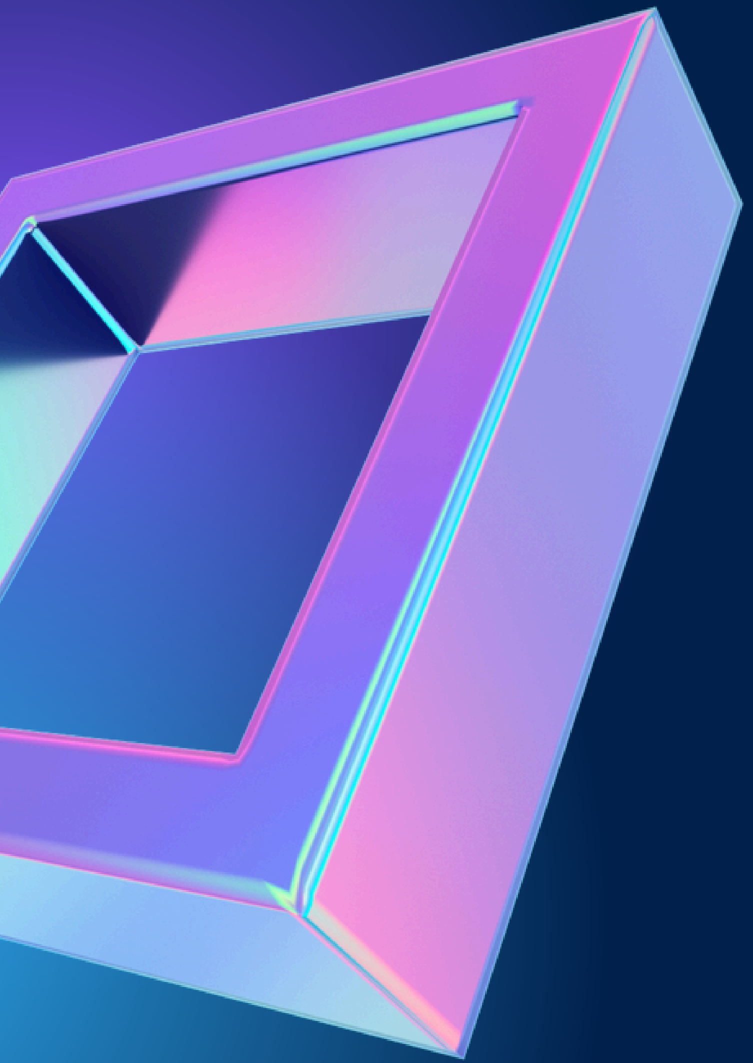
- MAE : 1013968.579587156
- RMSE: 1398115.6668385956
- R^2 Score: 0.6132752494107068

Conclusion

- Successfully developed a house price prediction system.
- Applied data cleaning, visualization, and machine learning techniques.
- Random Forest produced more accurate predictions.
- Machine learning can support accurate real-estate valuation.

Recommendation

Real-estate businesses should prioritize properties with larger living areas, modern amenities, adequate parking facilities, and desirable locations. Leveraging predictive analytics can improve pricing strategies, reduce valuation errors, and support data-driven decision-making for both buyers and sellers



Thank You